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ELECTRICITY GENERATION FROM THE ROTATIONAL SPEED OF TRAIN WHEEL: AN APPROACH OF FUTURE ENERGY

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Abstract*- The demand for energy is increasing consistently in the global environment. While the resources of energy are decreasing. Additionally, sources of energy create environmental pollution. To address this issue, the use of renewable sources of energy would be a potential solution for the global environment. However, renewable sources of energy mainly depend on environmental and geographical conditions. Hence, harvesting energy from existing sources can provide supplemental energy to meet the growing demand for energy. To this end, this paper presents a potential source for electricity generation using the rotational speed of an electric vehicle such as a train. This paper proposed a system, where electricity can be produced from the rotational speed of wheels of trains. The produced electricity can be used in several applications including to operate fans, lights, and other internal electrical devices on the train. It is found that the required amount of annual fuel costs will be reduced by about $35040. This paper also presents the total initial cost of our proposed system like the cost of the fabricated wheels, motor, battery, copper wire, rotor, and other accessories of the train. Our proposed system takes 750 days or 25 months to recover the initial cost. Besides, No generator is needed in the compartments of the train and thus it saves some space in the compartment for the passengers. The proposed system can be implemented to harvest energy from the rotational speed of wheels of trains that can help to make a pollution-free environment.*

Keywords: Electricity generation, rotational speed, future energy, less pollution, energy demand

1. INTRODUCTION

The energy crisis is a common phenomenon in recent years and the energy demand will increase very highly by one-third from 2010 to 2035 globally as presented in Figure 01 [1, 2]. On the other hand, the conventional resources of energy are decreasing that are the main sources of energy [3, 4]. These resources including coal, oil, and natural gas have become the main energy sources to mitigate this energy demand. The combustion of fossil fuel is one of the main energy sources which supplies about 85% of the world's energy supply [5]. These fossil fuels are burned by heavy industries and electric vehicles and release greenhouse gases such as CO_2 in the atmosphere at an alarming rate which increases global earth temperature [6]. In recent years two-third of the greenhouse effect was caused by human activities. From 1960 to 2010, the concentrations of CO_2 in the atmosphere have monotonically increased from 310 to 390 ppm [7]. Thus, developing countries are focusing on renewable energy sources such as biomass, solar, wind, and hydropower [1]. In a similar way, developing country like Bangladesh is dependent on fossil fuels, and the resources of oil, coal, and natural gas are too limited with respect to other countries [5]. In Bangladesh, the consumption of energy by fuel oil is 19%, coal is 2%, natural gas is 64%, high-speed diesel is 7%, renewable oil is 3% and imported power is 5% as depicted in Figure 02 [6]. Here, 98.5 percent of electricity generation is dependent on fossil fuels whereas renewable energy provides only 0.3 percent [1]. In Bangladesh, the consumption of energy by electric vehicles like trains from fossil fuels is a lot and produces greenhouse gases. Many developed countries are focusing on using renewable energies in electric vehicles to reduce the use of conventional fossil fuels and as well as energy demand [7].

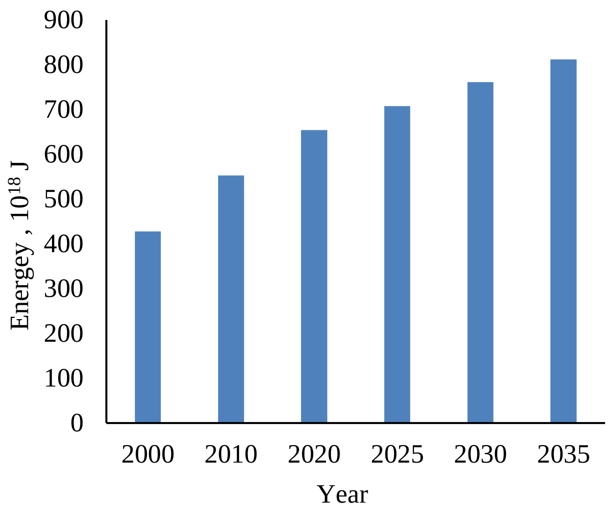

Fig. 01: Global energy demand is increasing with time [2]

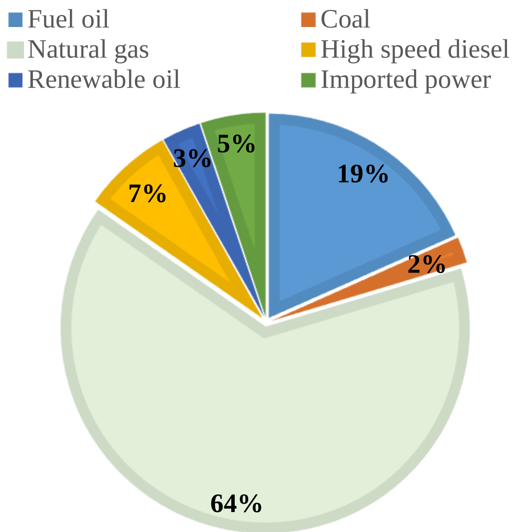

Fig. 02: Consumption of energy by different sources in Bangladesh [6]

In Bangladesh, all the engines of the trains are diesel locomotive engines. Besides, all the train uses diesel engine for the generation of electricity to give electrical support like coach lighting, air conditioning, laptop, and phone charging, refrigerating, and so on. Besides locomotion, the train needs a lot of energy to produce electricity for giving electrical support in all compartments. The cost of the fuel for the generation of electricity is very costly and not environmentally friendly.

Therefore, this paper proposed a system for the generation of electricity from the rotational speed of the wheels of a train which is rare in prior research, especially for Bangladesh. In this proposed system, a rotating disc (rotor) is attached to the wheel of the train by using a belt. Thus, when the wheels of the train will rotate, the rotating disc (rotor) can rotate. This rotating rotor can produce electricity with the help of the motor and the produced electricity is stored in the battery. . The produced electricity can be used in fans, lights, and other internal electrical devices in the train. As a result, no generator is required for the internal electrical devices in a train.

2. METHODOLOGY

2.1 Objective

The main objective of this paper is to produce electricity from the rotational speed of the train which can give power supply to the electrical devices of the components of the train. Thus, consumption of fuel can be reduced and as a consequence annual fuel cost can be reduced. Thus for electricity supply, large generator is not need. Besides, it can save some space for the passengers. Another objective of this paper is to ensure a pollution-free environment by reducing the emission of CO_2 and other toxic gases.

2.2 Methodology of Our Proposed System

While producing the current by using a turbine, the rotational energy is converted into electrical energy. In this research, a similar principle of electricity generation from the rotational speed of the turbine is used for the generation of electricity from the rotational speed of the wheels of a train. In this system, the kinetic energy obtained due to the forward movement of a train is converted into electrical energy. A motor is used as a generator for converting kinetic energy into electrical energy. The overall methodology for the generation of electricity is presented using a flow chart as shown in figure 03. The proposed system and its mechanism are presented through a developed model as presented in figure 04.

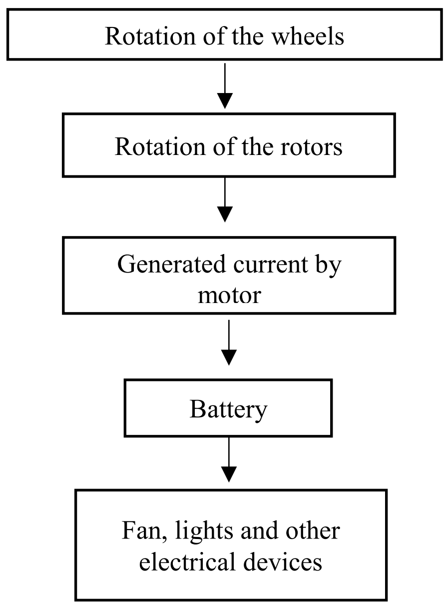

Fig. 03: Flow chart of proposed method

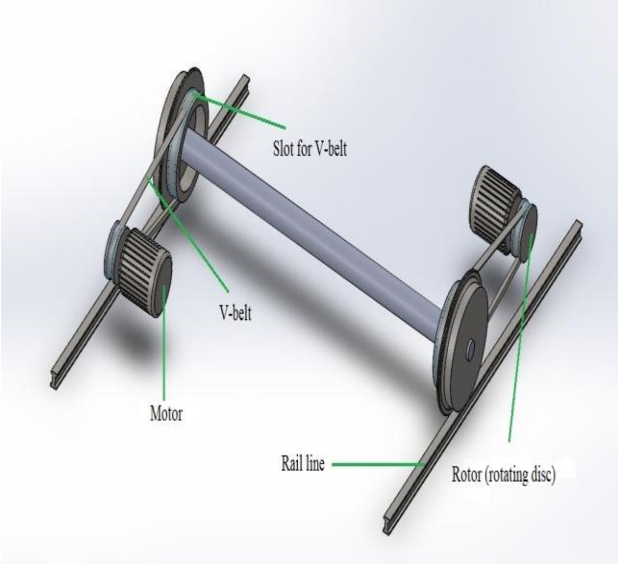

Fig. 04: Proposed method to produce electricity by using rotation of wheels

Firstly, it is considered that there are two slots in the wheels where one is larger and another is smaller. With a larger slot, the train moves through the line, while in the smaller slot, a V-belt remains to attach which is connected with the rotor (rotating disc). The rotor remains behind the wheel which can rotate with the rotation of the wheel of a train. As the circumference of the smaller slot of the wheel is three times larger than the rotor and the rotor rotates three times for a single rotation of the wheel. If the speed of the wheel is more, the rotation would be higher.

On the other side of the rotor, a motor presents which works as a generator to produce electricity. Due to the rotation of the rotor electromotive force (EMF) is induced in the stator winding of the motor by Faraday's law of electromagnetic induction. The electromotive force (EMF) produced by the motor is carried out by low voltage wires towards the batteries. The electromotive force (EMF) generated by the motor is in alternating current (AC) form but the battery stores direct current (DC). Therefore, an inverter is used to convert alternating current (AC) to direct current (DC). The batteries are used to supply current at a specific voltage and can work as an instant power supply (IPS). The supply current is used for running fans, lights, and other electrical devices of the compartments. Thus, no large generator is needed to meet up the required electricity demand. This can really help to reduce the initial production cost of a train as the generator is not needed which requires the high installation and operating costs. In this way, the proposed system can meet up the growing demand for energy.

Besides, this paper shows the cost of the diesel that has been saved by this proposed system which is needed for the electrical components of the compartments of the train. This paper also presents the total initial cost of our proposed system like the cost of the fabricated wheels, motor, battery, copper wire, rotor and other accessories. This paper also shows the time required to recover the cost of our proposed system.

3. Proposed System Design

In the proposed system all the components need to be in a proper operating condition such that the train would not face any kind of accident. In this system, a motor is used and due to the variation of the speed of the train a 2 pole motor is recommended as the sustainable speed of the motor is too high. The motor which is used at the bottom of the compartment needs to be exactly the upper side of the rail. The other components are a rotor which is just like the wheel of the train, transmission lines and in the proposed system a low voltage transmission line is used as the voltage produced by the motor is not so high. A copper wire can be used as a transmission line.

To store the generated direct current (DC) supply, a lead-acid battery is used. The major advantages of lead-acid batteries are high efficiency and life cycle, free maintenance, available in all shapes, and not harmful to nature.

In the case of the braking system, an electromagnetic braking system is used for the wheels of the motors. An electromagnetic braking system can control the speed of the wheels of the motors and thus we can reduce the speed as we desire. On the other hand, the cooling system for producing electricity, motors usually produce heat due to the friction between the copper brushes and winding. An induction motor can be used where the generation of heat is diminished as there is no friction between those two components. The heat produced in the rotor can be dissipated by air. In the transmission lines, heat is produced. However, if a low resistance wire such as copper wires is used then the heat generation can be avoided in electricity transmission lines.

3.1 Advantage of the Proposed System

The major advantages of the proposed system include that it can produce electricity all the time when the train is in moving condition, no need to establish any generator that reduces the number of compartments needed for the generator which eventually reduces the initial cost. In this proposed system, there is no effect on the speed of the train and thus train speed would not be reduced. Additionally, the overall maintenance cost is low and the efficiency of the system will be increased even with the use of a motor instead of a generator.

4. RESULTS AND DISCUSSION

In Bangladesh, a total of 348 trains carry passengers and goods and among them 86 trains are inter-city, 132 trains are mail express, 4 trains are international trains, and 126 trains are local trains. The average speed of the train is 43 km/hour [8].

4.1 Speed (RPM) of Analysis of Motor

As the average speed of the train is 43 km/hour, the train travels at an average speed of $\frac{43\times1000}{60\times60} = 11.94$ m/s. The diameter of the wheels of a train is about 1m and the circumference of the wheel is $2\times\pi\times\frac{1}{2} = 3.1416$ m. Hence, the wheel of the train gives $\frac{11.94}{3.1416} = 3.08$ rotation per second (RPS).

As the smaller slot is attached to the wheel of the train, it can give the same rotation per second (RPS). If the diameter of the smaller slot is about 0.75 m then it can also give the 3.08 rotation per second (RPS). If the diameter of the rotor is 0.250 m and it is attached with the slot with a belt then the rotor gives $\frac{0.75 \times 3.08}{0.250}$ = 9.24 rotations per second (RPS).

Therefore, the speed of the rotor is 9.24×60 = 554.4 rpm. Hence, the motor produces electricity with the help of rotor speed which is 554.4 rpm.

4.2 Analysis of Electricity Generation

The speed of the train is usually very high and low at different junctions. The speed of the 2 pole motor is about 3000 rpm which is very high. Hence, the proposed system uses a 2 pole motor. The torque of a 2 pole motor is about 2 N-m. While at 50 Hz frequency, 554.4 rpm, and 2 N-m torque the produced power by the motor can be determined by the following equation.

$$\text{Power} = \frac{2\pi NT}{60} \quad (1)$$

Where, N is rotational speed (554.4 rpm) and T is torque (2 N-m).

It is found that the generated electricity is 116.11 W for each wheel. In general, there are 8 wheels in one compartment of a train. The proposed system with the developed technology uses 8 wheels to generate a total of 929 W electricity from one compartment of a train. This electricity can be stored in a battery by converting through an inverter. The battery can supply the power to the electrical devices of the train.

In the train system of Bangladesh, there are 25 lights of 25 W and 20 fans of 32 W in one compartment. Therefore, the total power of fans and light = total power of 25 lights + total power of 20 fans with a total of 1.165 kW (= {(25×25) + (20×32)} W = 1.165 kW).

In 1 hour, the total consumption of power in one compartment is, W = P.t = 1.165×1 = 1.165 kWh. In 1 hour, total electricity produced by 8 wheels for one compartment is W = P.t = 0.929×1 = 0.929 kWh. The total deficit of electricity in 1 hour for one compartment is 1.165 - 0.929 = 0.236 kWh. The deficit of electricity is too much less than the produced electricity from the train. Thus, the wheel can give great support to the electrical devices like fan, light, water filter, phone, and laptop charging of the train.

4.3 Electricity Production with Different Sizes of Smaller Slot of Train Wheel and Rotor

For 0.75 m diameter of smaller slot and .25 diameter of rotor, the electricity production from 8 wheels for 1 compartment is 0.929 kWh. But the total consumption of power in 1 compartment is 1.165 kWh. So, there is deficiency of electricity. Thus, our proposed system had to produce more power than 1.165 kWh such that there is no deficiency of electricity. So, we tried to find out the required diameter of the smaller slot of the train wheel and the rotor such that there is no deficiency of electricity.

Table 01 describes that with the increase of the diameter of the smaller slot of the train wheel, the electricity production is increased. And with the increase of the diameter of the rotor the electricity production is decreased. At 0.70 m diameter of smaller slot with 0.15 m diameter of the rotor, 0.75 m diameter of smaller slot with 0.15 m diameter of the rotor, 0.80 m diameter of smaller slot with 0.15 m diameter of the rotor and 0.80 m diameter of smaller slot with 0.20 m diameter of the rotor there is no deficiency of electricity. So, after using these parameters in our proposed system there is no deficiency of electricity.

Table 01: Acceleration vs. frequency of vibration

Diameter of the Smaller Slot (m)	Diameter of the Rotor (m)	Speed of the Rotor (rpm)	Electricity Production from 1 Motor of 1 Wheel (watt)	Electricity Production for 1 Compartment (X) (kWh)	Deficiency of Electricity in 1 Compartment (1.165-X) (kWh)
0.70	0.15	862.2	180.57	1.445	No Deficit
	0.20	646.8	135.47	1.084	0.081
	0.25	517.2	108.32	0.867	0.298
	0.30	430.8	90.22	0.772	0.443
0.75	0.15	924	193.52	1.548	No Deficit
	0.20	693	145.14	1.161	0.004
	0.25	554.4	116.11	0.929	0.236
	0.30	462	96.76	0.774	0.391
0.80	0.15	985.2	206.34	1.651	No Deficit
	0.20	739.2	154.81	1.239	No Deficit
	0.25	591	123.77	0.990	0.175
	0.30	492.6	103.17	0.825	0.34

a) For 0.70 diameter of smaller slot:

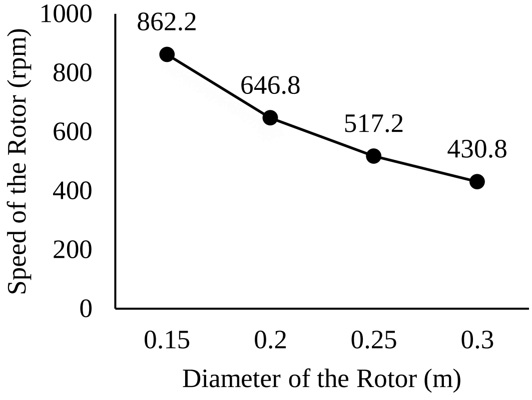

Fig. 5: Diameter of the rotor vs. Speed of the rotor

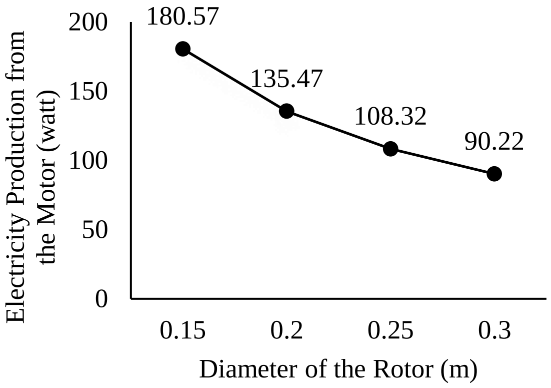

Fig. 6: Diameter of the rotor vs. Electricity production from the motor

b) For 0.75 m diameter of smaller slot:

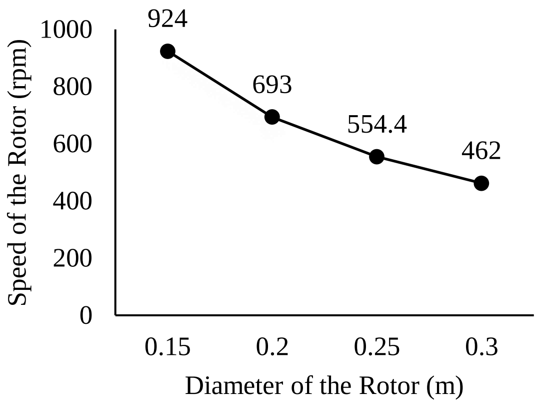

Fig. 7: Diameter of the rotor vs. Speed of the rotor

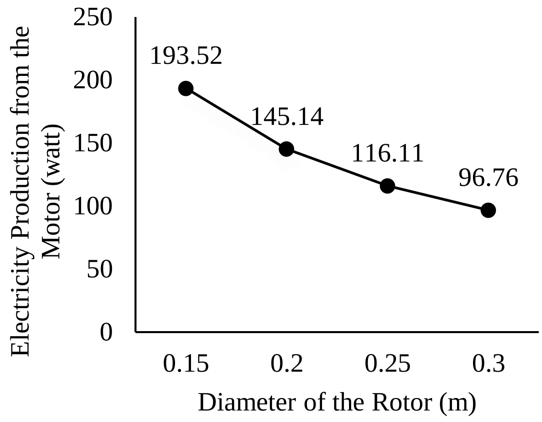

Fig. 8: Diameter of the rotor vs. Electricity production from the motor

c) For 0.75 m diameter of smaller slot:

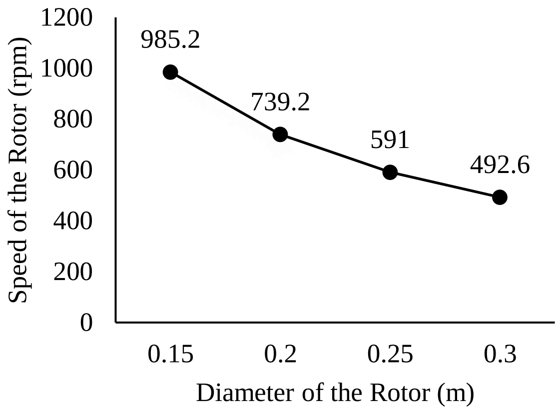

Fig. 9: Diameter of the rotor vs. Speed of the rotor

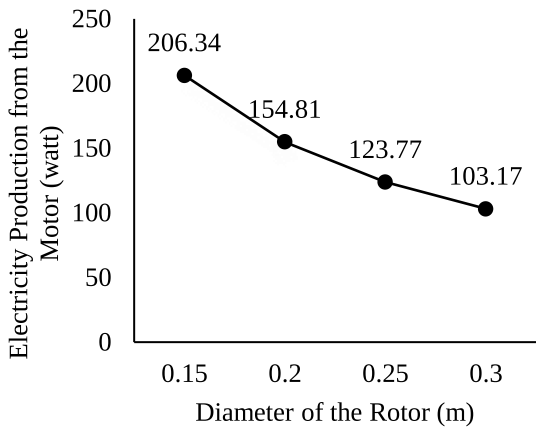

Fig. 10: Diameter of the rotor vs. Electricity production from the motor

So, with the increase of the diameter of the rotor the speed of the rotor is decreased and thus electricity production from the motor is also decreased.

4.4 Cost Analysis

In Bangladesh, two rating power generators have been used to generate electricity for train compartments. One is a 315 KVA 8 cylinder alternator and another is a 375 KVA 8 cylinder alternator. In 18 compartments of a train, it requires 6 liters of diesel per hour to generate electricity [9]. The average price of 1-liter diesel around the world is $1. If the train moves 16 hours per day, it consumes 96 liters of diesel. The price of 96 liters of diesel is approximately $96. In 1 year the total cost of diesel for a train is around $35040 (96×365). If we use our proposed system then it can save $35040 per year.

4.4.1 Initial Cost of the Proposed System

For 1 wheel of a train, it is needed a motor, lead battery, copper wire, V-belt, rotor, pulley, slot, and other accessories. The total cost for 1 wheel is around $500 and the total cost for 1 train is the cost of 1 wheel multiplied by the number of wheels in 1 compartment with a total number of compartments. In total the initial cost is $72000 (500 × 8 × 18).

4.4.2 Playback Cost Analysis

After the development of the proposed system, the initial cost of the proposed system is $72000. On the other hand, the annual fuel cost for one train is $35040.

If the proposed system is developed in a train, the time required for cost recovery is $\frac{365}{35040} \times 72000$ days = 750 days or 25 months.

Our proposed system will take 750 days or 2 months to recover the initial cost.

While moving, a train provides a number of rotations in the wheel. Hence, the rotor will rotate a number of times and this rotation can generate electricity through the use of the proposed system. The weight of the rotor is low with respect to the wheel and it will not cause any weight problem and the speed of the train will not be reduced. A V-belt is used between the rotor and the smaller slot of the wheel which can rotate the rotor very smoothly as V-belt is much efficient. A huge amount of electricity is produced that can be stored through the use of a battery which can be used for supply of electricity in different compartments of a train. No generator is needed and thus it can save some space in the compartment for the passengers. All the equipment of the proposed system is much lighter with respect to the total weight of the train. At last, it is evident that our proposed system reduces the fuel consumption cost of the train.

5. CONSLUSIONS

A system has been proposed for the generation of electricity from the rotational speed of the wheel of a train. The amount of electricity generation depends on the speed of a train. As the speed increased, the amount and efficiency of electricity generation are also increased. By applying the proposed system, fuel cost can be reduced, and generated electricity from the wheel can be used for different applications in different compartments of a train. Hence it decreases the losses and harvests energies of the wheels of a train. It is obtained that an annual fuel cost is reduced about $35040 per year. The proposed system is a pollution-free source of energy that will not emit any emissions including CO_2, NOx, SO_2, and other toxic gases. Finally, the proposed system provides enough power to the electrical devices such as coach lighting, air conditioning, laptop, and phone charging, refrigerating, etc. of the train.

6. ACKOWLEDGEMENT

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7. REFERENCES

[1] Ahmed S, Islam MT, Karim MA, Karim NM. Exploitation of renewable energy for sustainable development and overcoming power crisis in Bangladesh. Renewable Energy. 2014; 72:223-35.

[2] Ak, N. and A. Demirbas, Promising sources of energy in the near future. Energy Sources, Part A: Recovery, Utilization, and Environmental Effects, 2016. 38(12): p. 1730-1738.

[3] Hoque, M., F. Rashid, and M. Aziz, Gasification and Power Generation Characteristics of Rice Husk, Sawdust, and Coconut Shell Using a Fixed-Bed Downdraft Gasifier. Sustainability, 2021. 13(4): p. 2027.

[4] Das, B.K., M. Hasan, and F. Rashid, Optimal sizing of a grid-independent PV/diesel/pump-hydro hybrid system: A case study in Bangladesh. Sustainable Energy Technologies and Assessments, 2021. 44: p. 100997.

[5] Joarder, M. S. A., F. Rashid, and T. A. Joarder. "Solution to air pollution for removing CO2 and SO2 from flue gases: a prospective approach." In Proc. Int. Conf. Mech. Eng. Renew. Energy. 2019.

[6] Tushar, M.S.H.K., Joarder, M.S.A., Raj, R.F., Sarker, M.A. and Mahmud, M.R., 2019. Hydrothermal Carbonization and Hydrothermal Liquefaction of Water Hyacinth. In Proc. Int. Conf. Mech. Eng. Renew. Energy.

[7] Joarder, M. S. A., T. A. Joarder, and F. Rashid. "Safety at industrial working place by separating Carbon Dioxide Gas." In Proceedings of the International Conference on Mechanical, Industrial and Materials Engineering, 2019.

[8] Momen AAM, Nath SK, Mostafa SMG. Design and Implementation of Hybrid Power Generation System for Electric Loads of Train. 2019 5th International Conference on Advances in Electrical Engineering (ICAEE): IEEE; 2019. p. 85-90.

[9] Uddin MA, Mostafa S, Miad AMA. Electricity Generation from Train Wheel Axle. 2018 International Conference on Innovations in Science, Engineering and Technology (ICISET): IEEE; 2018. p. 455-8.

7. NOMENCLATURE

Symbol	Meaning	Unit
N	Rotational Speed	(rpm)
T	Torque	(N-m)